COMPUTER NETWORKS LAB

CCP-CNS2026
Section: V 14

FINAL REPORT

Problem 1: Multi-Site Enterprise Network
Apex Manufacturing

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GitHub: https://github.com/iamRohma/CNS2026-Problem1

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1. Introduction

Apex Manufacturing is a company with three departments: Sales, Engineering, and Admin. The company required a secure internal network that separates departmental traffic, provides automatic IP addressing, hosts an internal web server, and protects against unauthorized access attempts.

This project involved designing and implementing a complete enterprise network solution using Cisco Packet Tracer. The network was built with security as a priority, including VLAN segmentation, Access Control Lists, DHCP Snooping, attack simulation, and mitigation.

2. Network Design

2.1 Topology

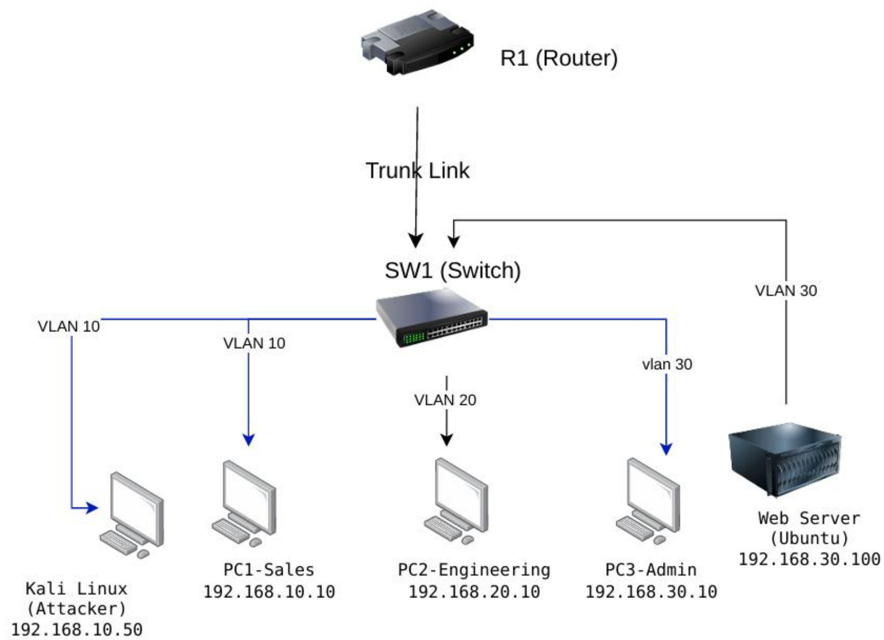
The network uses a Router-on-a-Stick topology with one Cisco 2911 Router and one Cisco 2960 Switch. All three departmental VLANs connect through the switch and communicate via the router sub-interfaces. This design is cost-effective and scalable for a small enterprise.

2.2 Devices Used

- Cisco Router 2911 (R1-ApexMfg) — Inter-VLAN routing, DHCP, ACLs
- Cisco Switch 2960 (SW1-ApexMfg) — VLAN segmentation, DHCP Snooping
- PC1-Sales (192.168.10.10) — Sales department workstation
- PC2-Engineering (192.168.20.10) — Engineering department workstation
- PC3-Admin (192.168.30.10) — Admin department workstation
- Web-Server (192.168.30.100) — Internal web server in Admin VLAN
- Kali-Attacker (192.168.10.50) — Simulated attacker in Sales VLAN

2.3 IP Addressing Plan

VLAN ID	Department	Network	Subnet Mask	Gateway	Device IPs
10	Sales	192.168.10.0	255.255.255.0	192.168.10.1	PC1: .10, Kali: .50
20	Engineering	192.168.20.0	255.255.255.0	192.168.20.1	PC2: .10
30	Admin	192.168.30.0	255.255.255.0	192.168.30.1	PC3: .10, Server: .100



3. Configuration

3.1 VLAN Configuration

Three VLANs were created on the Cisco 2960 switch to separate departmental traffic:

- VLAN 10 (Sales): Ports Fa0/2, Fa0/6
- VLAN 20 (Engineering): Port Fa0/3
- VLAN 30 (Admin): Ports Fa0/4, Fa0/5
- GigabitEthernet 0/1 configured as trunk port to carry all VLANs to the router

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Fa0/1, Fa0/7,
Fa0/8, Fa0/9
                                                Fa0/10,
Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14,
Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18,
Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22,
Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Sales                            active    Fa0/2, Fa0/6
20   Engineering                      active    Fa0/3
30   Admin                            active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
Switch#
Switch#
```

3.2 Router-on-a-Stick Configuration

The router uses sub-interfaces on GigabitEthernet0/0 to route traffic between VLANs:

- Gi0/0.10 — encapsulation dot1Q 10, IP 192.168.10.1/24
- Gi0/0.20 — encapsulation dot1Q 20, IP 192.168.20.1/24
- Gi0/0.30 — encapsulation dot1Q 30, IP 192.168.30.1/24

```
Router#
Router#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  up
up
GigabitEthernet0/0.10  192.168.10.1    YES manual up
up
GigabitEthernet0/0.20  192.168.20.1    YES manual up
up
GigabitEthernet0/0.30  192.168.30.1    YES manual up
up
GigabitEthernet0/1     unassigned      YES unset
administratively down down
GigabitEthernet0/2     unassigned      YES unset
administratively down down
Vlan1                  unassigned      YES unset
administratively down down
Router#
Router#
```

3.3 DHCP Configuration

The router was configured to provide automatic IP addressing for all three VLANs using three DHCP pools: SALES, ENGINEERING, and ADMIN. Each pool was assigned the correct network, subnet mask, and default gateway.

3.4 DHCP Snooping

DHCP Snooping was enabled on the switch to prevent rogue DHCP servers from assigning incorrect IP addresses to clients. Only the trunk port (Gi0/1) connected to the router was marked as trusted.

3.5 Web Server

An HTTP web server was configured on the Web-Server device (192.168.30.100) in the Admin VLAN. The server hosts an internal webpage accessible to all departments except the blocked Sales VLAN.

4. Security Implementation

4.1 Access Control List (ACL)

A combined ACL named SALES_SECURITY was configured on the router's Gi0/0.10 sub-interface:

- Rule 1 — Block Kali attacker (192.168.10.50) from accessing any destination
- Rule 2 — Block entire Sales VLAN from accessing Admin VLAN
- Rule 3 — Permit all other traffic

```
Router#show access-lists
Extended IP access list SALES_SECURITY
    10 deny ip host 192.168.10.50 any
    20 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
    30 permit ip any any

Router#show access-lists
Extended IP access list SALES_SECURITY
    10 deny ip host 192.168.10.50 any
    20 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
    30 permit ip any any

Router#
```

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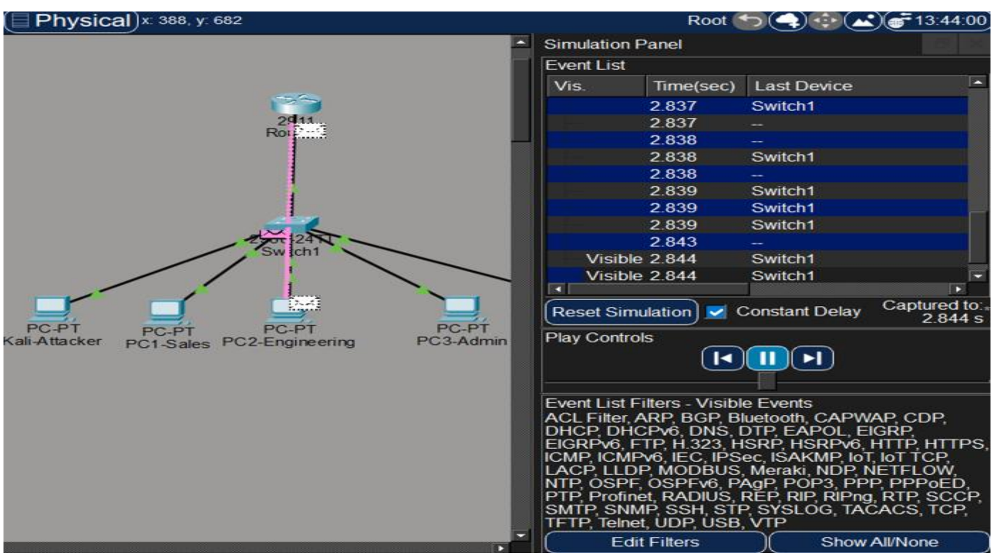
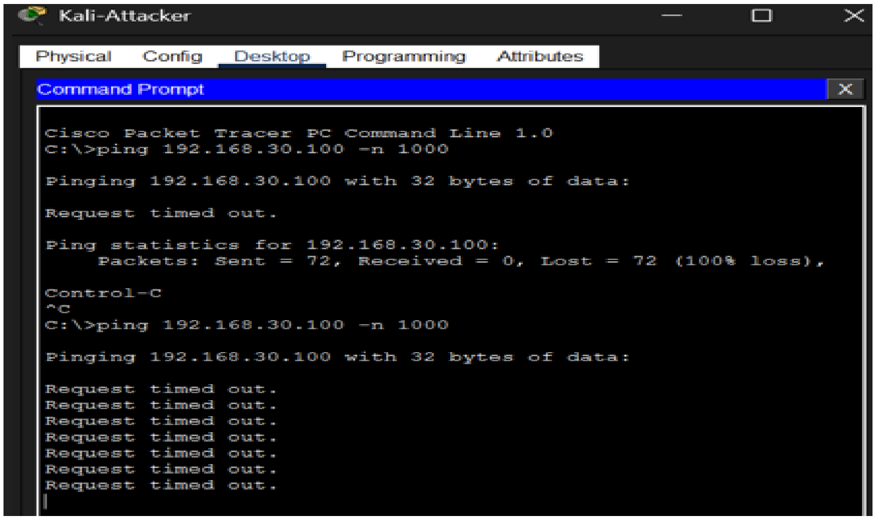
4.2 Attack Simulation

A DoS attack was simulated from Kali-Attacker (192.168.10.50) targeting the web server (192.168.30.100) using a ping flood of 1000 ICMP packets. The attack was visualized using Packet Tracer Simulation Mode.



4.3 Attack Mitigation

The SALES_SECURITY ACL successfully blocked all 1000 attack packets with 100% packet loss on the attacker side, confirming the mitigation was effective.



4.4 Security Test Results

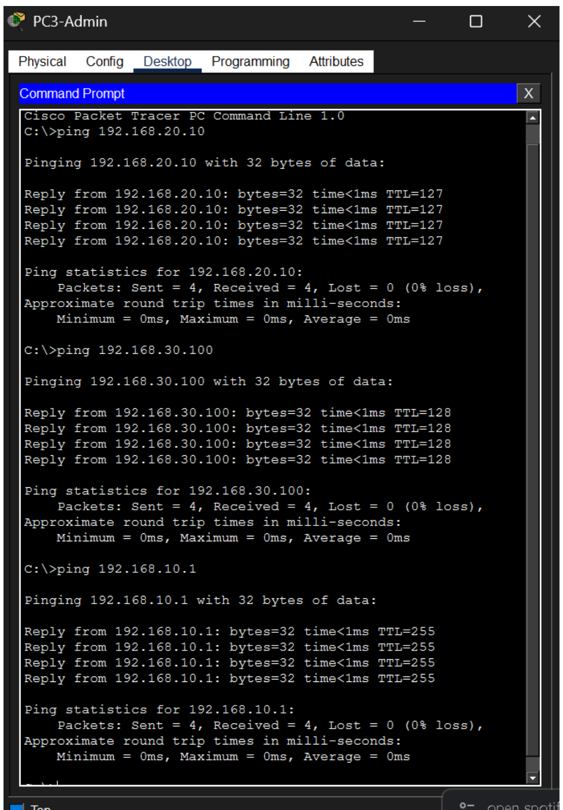
Test	Source	Destination	Expected	Result
ACL Block	PC1-Sales	PC3-Admin	BLOCKED	PASS
ACL Block	Kali-Attacker	Web-Server	BLOCKED	PASS
ACL Allow	PC2-Engineering	PC3-Admin	ALLOWED	PASS
ACL Allow	PC2-Engineering	Web-Server	ALLOWED	PASS
DoS Attack	Kali-Attacker	Web-Server	BLOCKED	PASS

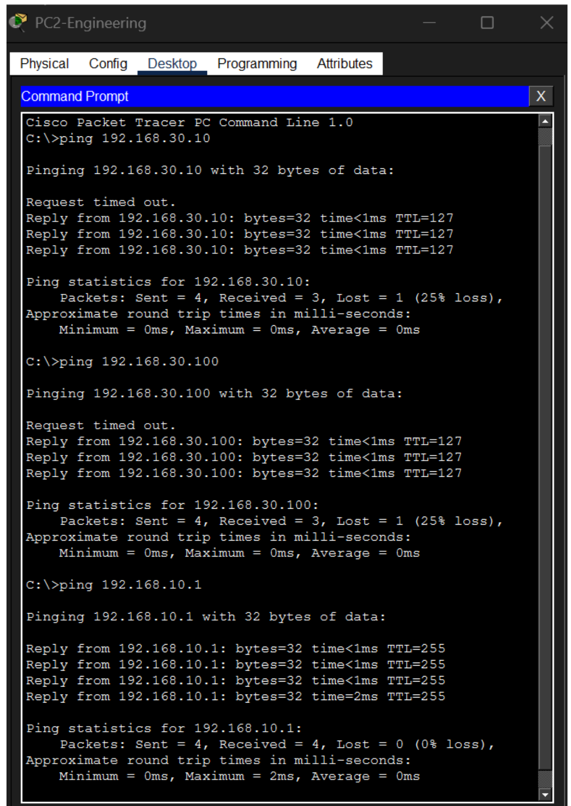
5. Testing & Conclusion

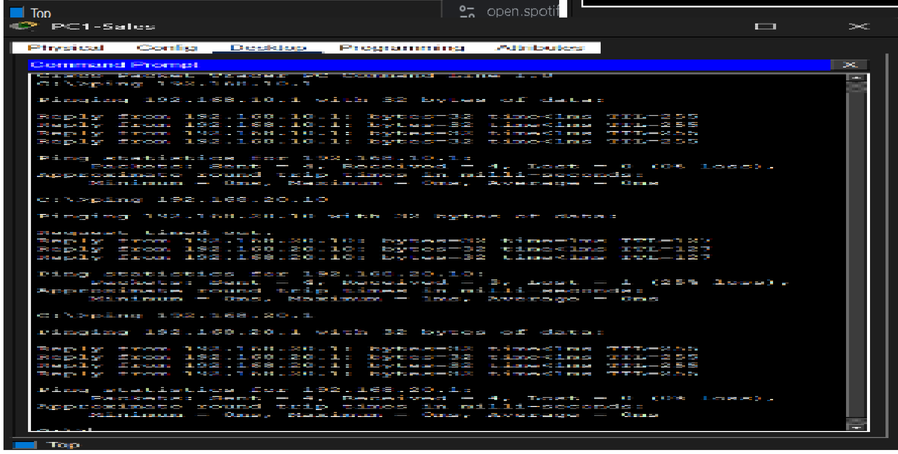
5.1 Ping Test Matrix

From	To	IP Address	Result
PC1-Sales	Router Gateway	192.168.10.1	PASS

PC1-Sales	PC2-Engineering	192.168.20.10	PASS
PC1-Sales	PC3-Admin	192.168.30.10	BLOCKED
PC1-Sales	Web-Server	192.168.30.100	BLOCKED
PC2-Engineering	PC3-Admin	192.168.30.10	PASS
PC2-Engineering	Web-Server	192.168.30.100	PASS
PC3-Admin	PC2-Engineering	192.168.20.10	PASS
Kali-Attacker	Web-Server	192.168.30.100	BLOCKED







5.2 Conclusion

This project successfully implemented a secure enterprise network for Apex Manufacturing. All required objectives were achieved:

- Three VLANs created and verified for department isolation
- Inter-VLAN routing working correctly via Router-on-a-Stick
- DHCP providing automatic IP addressing to all devices
- DHCP Snooping blocking rogue DHCP servers
- ACL blocking Sales VLAN from Admin VLAN
- DoS attack simulated and successfully mitigated
- Web server accessible to authorized departments only

5.3 Tools Used

- Cisco Packet Tracer — Network simulation and configuration
- Wireshark — Packet capture and analysis
- draw.io — Network topology diagram
- GitHub — Version control and file submission